Automata for MSO over Infinite Trees with Quantification over Borel Sets of Branches

Mikołaj Bojańczyk, Antonio Casares, Sven Manthe and Paweł Parys

Abstract

Rabin's Tree Theorem says that the MSO theory of the infinite binary tree 2^* is decidable. Shelah showed that MSO logic becomes undecidable if this tree is extended to $2^{\leq\omega}$, i.e. by allowing quantification over sets of infinite branches. A longstanding open problem is whether the decidability can be recovered in $2^{\leq\omega}$ by restricting set quantification to Borel sets. We make some progress in this direction, by identifying a suitable automaton model, and showing that most of the automata-theoretic approach to Rabin's Theorem can be extended to the new framework. The only missing part is a conjecture about finite memory determinacy in certain games. This paper states and explores the conjecture. We prove it in some restricted cases, and give lower bounds on the memory required in those games.

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1 Introduction

One of the landmark results of logic in computer science is Rabin's Tree Theorem [5], which says that the MSO theory is decidable for the infinite binary tree, which is the set 2^* equipped with the prefix relation. This is a powerful theory, and it is close to the border of undecidability. On the other side of this border we find the Cantor set 2^ω, equipped with the lexicographic order, and also the order of the reals. Both of these structures have an undecidable MSO theory (as shown by Shelah [6, Theorem 7], assuming the Continuum Hypothesis; this assumption was later removed in [3], i.e., undecidability can be proved in ZFC only). The undecidability proofs use sets that are complicated from a topological point of view, and Shelah conjectured [6, Conjectures 7A and 7B] that decidability is recovered once one restricts set quantification to range over Borel sets (we use the name *Borel MSO* for the resulting logic):

A. The Borel MSO theory is decidable for $(2^{\leq\omega}, \preceq, \leq)$, where $\preceq$ is the prefix order and $\leq$ is the lexicographic order.
B. The Borel MSO theory is decidable for $(\mathbb{R}, \leq)$.

Conjecture B is the weaker one, since the real numbers can be defined using Borel MSO in the first structure. The weaker conjecture was recently proved by Manthe [].

This paper is devoted to the stronger Conjecture A. One can think of this conjecture as being a common generalisation of Rabin's Tree Theorem and Conjecture B. This is because the complete binary tree is recovered in $2^{\leq\omega}$ as the definable substructure consisting only of nodes in 2^*. Hence, any proof of Conjecture A would also need to subsume a proof of Rabin's Tree Theorem. In particular, this seems to rule out the proof of Manthe, which is based on linear orders and the compositionality of their MSO theories, and does not seem to be amenable to extensions that cover the tree structure. This paper proposes a programme to prove Conjecture A. Our main contributions can be summarised as follows:

(1) We introduce a certain kind of infinite games, and conjecture that these games can be won with finite memory.
(2) We prove that if the conjecture on finite memory is true, then the Conjecture A is also true, that is, one can decide the Borel MSO theory of $2^{\leq\omega}$, and furthermore the logic has a corresponding automaton model.
(3) We present a preliminary study of the conjecture on finite memory, with some partial positive results.

Let us describe these contributions in some more detail. Simply put, our programme is to copy the proof of Rabin's Tree Theorem, and extend it to take into account Borel sets of branches. We begin by briefly recalling how Rabin's Tree Theorem is proved, using a modern presentation that emphasises games, as proposed by Gurevich and Harrington [2]. In the proof, one shows that every MSO formula can be transformed into an equivalent automaton, and the latter can be tested for emptiness. To prove this, one needs to show that automata have the same closure properties as the logic, namely Boolean combinations and projection. Union and intersection are easily achieved using a product construction, and projection is also easy because the automaton model in question is nondeterministic. The difficult closure property is complementation, and unlike for ω-words, determinisation is not an option. To prove complementation, one describes rejection by the automaton in terms of a game of infinite duration, and then proves that this game can be won using simple strategies of finite memory. The finite memory can then be embedded into the state space of a nondeterministic automaton which recognises the complementary language.

As we describe in this paper, most of the proof described in the previous paragraph can be adapted to the more general setting of $2^{\leq\omega}$. In the more general setting, an input tree has type $2^{\leq\omega} \to \Sigma$, that is, it assigns labels to both nodes and branches, and a run of the automaton has type $2^{\leq\omega} \to Q$. In both cases, the maps are required to be Borel, because on the side of logic we are targeting the Borel MSO. The acceptane condition for the automaton is a variant of the Muller condition, and takes into account the states assigned to the branches. Using this automaton model, we are able to implement all parts of the proof of Rabin's Tree Theorem, with one exception, namely finite memory determinacy of the underlying games. This

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determinacy result is the content of our conjecture, and much of the paper is devoted to studying it.

The conjecture. Let us now describe in more detail our conjecture on finite memory strategies. Like most games studied in the context of automata, such as parity games or Muller games, our games are infinite duration games played by two players, called Adam and Eve, using directed graphs (called arenas), where each position is owned by one of the two players. This notion of game is standard, and will be recalled later in the paper. In this introduction, we focus on the distinguishing features of the games that arise within the context of Borel MSO, which are defined below. We call the games Muller-Borel, because the acceptance condition is a Boolean combination of Büchi and Borel conditions, and Boolean combinations of Büchi conditions are called Muller conditions.

Definition 1.1. *A* Muller-Borel game *is a game where:*

(1) ***Tree-like arena.*** *The set of positions is $2^* \times Q$ for some finite set Q. Edges of the game can only move to child nodes, that is, if*

$$(v, q) \to (w, p)$$

is an edge in the game, then w is a child of v. Not all edges that move to children need to be available.

(2) ***Muller-Borel condition.*** *The winning condition is a finite Boolean combination of two kinds of conditions:*
(Büchi) *$q \in Q$ is seen infinitely often;*
(Borel) *the branch belongs to a Borel set $B \subseteq 2^\omega$.*

Using Martin's Borel Determinacy Theorem [4, p. 371], one can show that the games are determined, that is, one of the players must have a winning strategy. However, our main interst is in the memory used by the winning strategy, which is beyond the scope of Martin's Theorem. Our conjecture is that the games can be won with finite memory, as stated below. (The notion of finite memory is the standard one, and will be recalled later in the paper.)

Conjecture 1.2. *For every Morel-Borel game, one of the players has a winning strategy that uses finite memory. Furthermore, an upper bound on the memory size can be computed based on the size of Q.*

Using the same framework as for Rabin's Tree Theorem, we prove in Theorem 4.6 that the above conjecture implies decidability of Borel MSO on $2^{\leq\omega}$, that is, Shelah'a Conjecture A. The rest of this introduction is devoted to a preliminary discussion of the conjecture, with an extended discussion contained in the main body of the paper.

In the first condition on a tree-like arena, the reader will recognise the kind of games that appear in the analysis of tree automata, with Q being the set of states. For many kinds of games, such as Muller games or parity games, the tree-like shape of an arena is irrelevant in the analysis of the game, and hence can be ignored in the definition of the game. In our case, however, the tree-like shape is essential, since the Borel conditions in the game are defined in terms of the tree structure. Let us now explain how certain special cases of a Muller-Borel game correspond to important results on determinacy of games.

- Suppose that we only allow Büchi conditions in the game. In this case, we have a game with a Muller condition. Muller games are known to have finite memory strategies, which was first proved by Büchi and Landweber for finite arenas [1, Theorem 1'], and then by Gurevich and Harrington for general arenas [2, Theorem 1]. Therefore, if we only allow Büchi conditions, then the conjecture is true.
- Suppose that there is only one state in the game. In this case, each game position can be reached in only one way (since the positions form a tree), and therefore all strategies are memoryless. Thus, from Martin's Theorem about determinacy (TODO: I'm fine with either option, but currently we use at least 3 different names for this theorem in the paper) we can conclude that the conjecture holds in this special case, even without any memory. Note that the full power of Martin's Theorem is needed for this special case.

Later on in the paper, we prove the conjecture for other special cases of the game, including the special case where only the Borel condition is allowed (which extends the situation from the second item above, in which the Büchi condition trivializes).

Our conjecture talks about finite memory strategies. A natural question would be about memoryless strategies. For example, in automata theory, finite determinacy of Muller games is currently proved by first reducing to a parity condition, and then proving that parity games can be won using memoryless strategies. The only need for memory arises from the reduction from Muller to parity, which is typically accomplished using a variant of McNaughton's "last appearance record" construction. For our conjecture, we are unable to present such a reduction, that is, we are not aware of a variant of our conjecture which would talk about memoryless strategies. Using priorities does not seem to help: even if we convert the game into a parity-style format where the Büchi part of the acceptance condition is replaced by a priority function, then memory is needed. In fact, even in the presence of only three priorities, the amount of needed memory grows unboundedly as a function of the number of states (see Proposition 5.12). On the positive side, we show that for two priorities, only two states of memory are needed, regardless of the number states (see Proposition 5.7).

The conjecture seems rather strong. In particular, as we discuss in Section 4.3, the conjecture implies that if we take Borel MSO, and we further restrict the semantics to use only Boolean combinations of F_σ sets, then the same formulas are true. In other words, Borel MSO can essentially only talk about Boolean combinations of F_σ sets, and it cannot access more complicated Borel sets. Since one could expect Borel MSO to be more expressive, this could be seen as evidence against the conjecture. However, the recent proof of Manthe seems to be consistent with our conjecture, since it implies that in the weaker setting of 2^ω one can also restrict quantification to Boolean combinations of F_σ sets, and the same formulas are true.

2 Borel MSO

In this section, we describe the logics in question. We begin with the usual notion of monadic second-order logic (MSO). In this logic, there are two kinds of variables: lowercase variables x, y, z for elements and uppercase variables X, Y, Z for sets. Formulas are built using

the following constructors:

$$\underbrace{\exists x\ \forall x}_{\text{quantification over elements}} \quad \underbrace{\exists X\ \forall X}_{\text{quantification over sets}} \quad \underbrace{\vee\ \wedge\ \neg}_{\text{Boolean combinations}} \quad \underbrace{R(x_1,\ldots,x_n)}_{\text{relations from the vocabulary}}.$$

The semantics of the logic are defined in the usual way (see e.g. [7] for a more in-depth discussion). When we talk about the MSO theory of a structure, we mean the set of MSO sentences that are true in this structure. Rabin's Tree Theorem [5] states that the MSO theory is decidable for the infinite binary tree, which is viewed as a structure where the underlying set is 2^*, and there are two binary relations: the prefix order, and the lexicographic order.

The logic that is studied in this paper is Borel MSO. This logic has the same syntax as MSO, except that the semantics are changed so that set quantification is restricted to range only over Borel sets. Recall that the family of Borel sets in a topological space is the least family of subsets that contains the open sets, and which is closed under complementation as well as countable unions and intersections. Therefore, in order to speak of Borel MSO, the underlying set in a structure must be equipped with a topology. The case which interests us is where the underlying set is $2^{\leq\omega}$, that is, strings of bits that have length at most ω. The finite strings are called *nodes* and the infinite ones are called *branches*. The topology is the one which arises from the metric defined for $x \neq y$ by the formula

$$d(x,y) = \frac{1}{\text{length of the longest common prefix of } x \text{ and } y}.$$

When we speak of $2^{\leq\omega}$ as a structure, we assume that it is equipped with the prefix and lexicographic orders, just as in Rabin's Tree Theorem. The prefix order gives us the tree structure on nodes, and it allows us to check if some node belongs to a branch.

This paper is devoted to Conjecture A from the introduction, which says that the Borel MSO theory of $2^{\leq\omega}$ is decidable, meaning that there is an algorithm that inputs a sentence of the logic, and answers if the sentence is true in the structure. Note that the Borel restriction is relevant only for branches. This is because every set of nodes is Borel, and thus a subset of $2^{\leq\omega}$ is Borel if and only if its restriction to branches is a Borel set.

Example 2.1 (Determinacy). In this example, we illustrate the difference between MSO and Borel MSO. A Gale-Stewart game is a game where two players, call them Adam and Eve, pick bits from $\{0,1\}$ in alternation. After ω rounds, an infinite play in 2^ω arises. The winner of the game is given by a winning condition $W \subseteq \{0,1\}^\omega$, which tells us which plays are winning for Eve. A strategy for player Eve can be described as a set of nodes X, which indicates the chosen successors used by the strategy. Similarly, a strategy for Adam can be described a set of nodes Y. Therefore, one can express determinacy of such games using a formula:

$$\forall W \subseteq 2^\omega.\ \vee \begin{cases} \exists X \subseteq 2^*.\ \forall Y \subseteq 2^*.\ \varphi(W,X,Y) \\ \exists Y \subseteq 2^*.\ \forall X \subseteq 2^*.\ \neg\varphi(W,X,Y), \end{cases}$$

where $\varphi(W,X,Y)$ says that the unique play consistent with strategies X and Y belongs to the set W. If we think of this formula as a formula of Borel MSO, then it will be true, since only Borel winning conditions will be considered, and Martin's Theorem says that such games are determined. If we think of this formula as a formula of (unrestricted) MSO, then it will be false, since there exist games with (non-Borel) winning conditions that are not determined, assuming the Axiom of Choice.

3 Nondeterministic automata over Borel trees

The idea behind the automata method is to find an automaton model that recognises exactly the same languages as those that can be defined in Borel MSO. These languages will contain trees, and the trees will be labelled. The intuitive idea is that the labels describe a valuation of free variables for a formula that has free variables. Let us begin by describing the appropriate notion of trees.

Definition 3.1 (Borel tree)**.** *A* Borel tree *over a finite alphabet Σ is defined to be a function*

$$t : 2^{\leq\omega} \to \Sigma$$

such that for every letter $a \in \Sigma$, the inverse image is a Borel set. We denote the set of Borel trees over the alphabet Σ by $\mathsf{T}\Sigma$.

Since the topology is only important on the infinite branches, the Borel condition is only important for those, and the labels of the nodes are unconstrained.

As mentioned earlier in this section, we will consider languages of Borel trees that can be defined in logic, or recognised by automata. Let us first explain how a language of Borel trees can be defined in logic. We say that a language $L \subseteq \mathsf{T}\{a_1,\ldots,a_n\}$ is definable in MSO if there is a formula

$$\varphi(X_1,\ldots,X_n)$$

of Borel MSO such that a tree $t \in \mathsf{T}\Sigma$ belongs to L if and only if the formula is true under the valuation that maps each variable X_i to the set of nodes and branches that are labelled by the letter a_i. Our goal in this section is to define an automaton model which recognises exactly the languages definable in Borel MSO. (This goal will be achieved conditionally on a certain conjecture about complementation of automata.)

Automata over Borel trees. We now describe the automaton model that will be used to recognise languages of Borel trees. This automaton is a variant of nondeterministic Muller tree automata.

Definition 3.2 (Nondeterministic automaton)**.** *A* nondeterministic Muller automaton on Borel trees *is given by:*

- *a finite set of states Q;*
- *a finite alphabet Σ;*
- *a subset $I \subseteq Q$ of initial states;*
- *two transition relations:*

$$\Delta_* \subseteq Q \times \Sigma \times Q \times Q \qquad \Delta_\omega \subseteq \mathsf{P}Q \times \Sigma \times Q.$$

An input Borel tree $t \in \mathsf{T}\Sigma$ is accepted by the automaton if there exists an accepting run, which is defined to be a Borel tree $\rho \in \mathsf{T}Q$ such that the following three conditions hold:

(1) the root is labelled by an initial state;
(2) for every node $x \in 2^*$, the transition relation Δ_* contains the tuple consisting of: the state in x, the label of x in t, the state in the left child of x, the state in the right child of x;
(3) for every branch $b \in 2^\omega$, the transition relation Δ_ω contains the tuple consisting of: the set of states that occur infinitely often along b, the label of b in t, the state in b.

Lemma 3.3. *Emptiness is decidable for nondeterministic Muller automata on Borel trees.*

Proof. We reduce the problem to emptiness for nondeterministic Muller automata on the usual kind of infinite trees, which do not have labels or states on branches. For the latter kind of automata, emptiness is decidable.

Consider a nondeterministic Muller automaton on Borel trees. The Muller automaton on Borel trees is nonempty if and only if there is a labelling $\rho : 2^* \to Q$ which satisfies the three conditions (1), (2), (3) in the definition of an accepting run, except that condition (3) is changed so that for each branch $b \in 2^\omega$, the set of states that occur infinitely often along b belongs to

$$\mathcal{F} = \left\{ R \subseteq Q \mid \ (R, a, q) \in \Delta_\omega \text{ for some } a \in \Sigma \text{ and } q \in Q \ \right\}.$$

We define a Muller tree automaton on infinite trees in the usual sense, which has $\mathcal{F}$ as its acceptance condition. This new automaton is nonempty if and only if the original Borel tree automaton is nonempty. □

We will now establish some straightforward closure properties of our automata, namely union, intersection, and projection. The difficult part will be complementation. We are unable to prove complementation, but the rest of this paper is devoted to a discussion of this problem, and some possible approaches towards its solution.

Let us begin with union and intersection.

Lemma 3.4. *Languages recognised by nondeterministic Muller automata on Borel trees are closed under union and intersection.*

Proof. Standard product construction. This construction works because the Borel condition for trees is closed under pairing. This means that if we take two trees over alphabets Σ_1 and Σ_2 and pair them into a single tree over the alphabet $\Sigma_1 \times \Sigma_2$, then the new tree satisfies the Borel condition if and only if both of the original trees did so. □

We now turn to projection, which corresponds to existential set quantification in logic. Since our automaton model is nondeterministic, it will also be closed under taking projections, understood as follows. Consider a function

$$f : \Sigma \to \Gamma$$

from one finite alphabet to another. This function can be extended to a function on Borel trees

$$\mathsf{T}f : \mathsf{T}\Sigma \to \mathsf{T}\Gamma$$

in the obvious way, by applying f to the label of every node. We can apply this function to some language $L \subseteq \mathsf{T}\Sigma$, giving a new language $f(L) \subseteq \mathsf{T}\Gamma$, which is called the *projection* of L along f.

Lemma 3.5. *Languages recognised by nondeterministic Muller automata on Borel trees are closed under taking projections.*

Proof. This is the same proof as usual: take the automaton before the projection, and apply the projection function to each input letter. This proof works because if we take the projection of a Borel tree, then the new tree is also Borel. (It is important to avoid confusion here: individual Borel trees are closed under projection, but Borel languages of Borel trees are not.) □

As in the proofs of Büchi or Rabin, the essential step is complementation. We do not know how to prove this, and leave it as a conjecture. In the next section, we will present a conjecture on finite memory strategies in games, which will imply that languages recognised by nondeterministic Muller automata on Borel trees are effectively closed under complementation, i.e. for every automaton one can compute an automaton for its complement language. This in turn will imply our main conjecture about decidability of Borel MSO.

4 Complementation of automata and games

When proving closure under complementation for nondeterministic automata on infinite trees, games and their determinacy play a central role. We follow that same path. Let us begin by recalling the kind of games that will be used in our proofs.

4.1 Games

The games that we consider have infinite duration, and are played by two players Adam and Eve on a directed graph. We now recall the definition of such games, and the basic concepts such as strategies.

Here, a directed graph is a set of vertices V together with a binary relation E on the vertices. We typically write $v \to w$ to indicate that there is an edge from v to w. Paths, both finite and infinite, are defined in the usual way.

Definition 4.1 (Game). *A game is given by:*

(1) a directed graph (V, E), called the arena*;*
(2) an ownership function of type $V \to \{\text{Adam}, \text{Eve}\}$;
(3) an initial position $v_0 \in V$;
(4) a winning condition W, which assigns a winner in $\{\text{Adam}, \text{Eve}\}$ to every infinite path starting in v_0.

In the context of a game, vertices of the underlying graph will be called *positions*. We will assume that in the arena, every position has at least one outgoing edge; this can be easily achieved by possibly adding two losing sink positions, one for each player.

We now briefly recall the usual notions of strategies and winning strategies for games. A *strategy* for player i is a function that inputs a finite path in the arena from the initial position to some position owned by i, and outputs an edge that extends this path. Once both players fix their strategies, a unique infinite path arises in the game, and the winning condition tells us which of the two players wins. A player wins if they have a winning strategy, i.e. a strategy which wins against every strategy of the opponent. A game is *determined* if one of the players has a winning strategy. Under the axiom of choice, not every game is determined, but Martin's Borel Determinacy Theorem says that the game will be determined if the winning condition is a Borel subset of all paths, according to the usual topology.

We will be particularly interested in *finite memory strategies*, which are defined as follows. Essentially, this is a strategy that is produced by a finite state transducer that reads the edges in the game. More formally, a strategy for Eve (for Adam, the concept is defined symmetrically) with memory M consists of some initial

memory state m_0 and two functions

$$\underbrace{\delta : M \times E \to M}_{\text{memory update}} \quad \text{and} \quad \underbrace{\sigma : M \times \overbrace{V_{\text{Eve}}}^{\text{positions owned by Eve}} \to E}_{\text{Eve's choice}} .$$

The first function is used to update the memory along a play (as in a deterministic automaton) and the second function shows how Eve chooses edges in the game from her positions. A *finite memory strategy* is one where the memory set M is finite. A game has *finite memory determinacy* if one of the two players can win using a strategy with finite memory. If the winning condition is a Muller condition, which topologically speaking is a finite Boolean combination of F_σ-sets, then it will have *finite memory determinacy*, i.e. the winning player can win using finite memory [2, Theorem 1]. A memoryless strategy is the special case of a strategy with finite memory in which the set M has one element only.

The two determinacy results discussed above are incomparable. Borel determinacy has a weaker assumption (the winning condition is Borel), but a weaker conclusion (the winning strategy might use infinite memory). Both results will play an important in our setting, since our games will be a hybrid of Borel and Muller games.

4.2 Complementation

To complement nondeterministic automata, we will use a game based approach, following Gurevich and Harrington [2]. In this approach, for a nondeterministic automaton $\mathcal{A}$ and an input tree t, we define an *acceptance game* as follows. In the game, Eve tries to show that the tree is accepted, by constructing an accepting run, and Adam tries to dispute Eve's claim, by showing that her run violates the condition Δ_ω. The positions of the game are pairs

$$\underbrace{(Q + \Delta_*)}_{\substack{\text{disjoint union of} \\ \text{states and transitions}}} \times 2^* .$$

The initial position consists of the initial state and the root of the tree. (One can assume without loss of generality that the automaton has one initial state.) In a position (q, v) that uses a state $q \in Q$, Eve chooses a transition from $(q, a, q_0, q_1) \in \Delta_*$ which has source state q. In the resulting position, Adam picks child $i \in \{0, 1\}$ and the game moves a position of the form $(q_i, v \cdot v_i)$. The winning condition in the game is defined as follows. Suppose that $P \subseteq Q$ is the set of states that are visited infinitely often in the play, and $b \in 2^\omega$ is the branch that is traced by this play in the underlying tree. Then Eve wins the play if and only if the branch b belongs to the set

$$\left\{ b \in 2^\omega \mid \text{ there is a state } q \text{ such that } (P, t(b), q) \in \Delta_\omega \right\} . \quad (1)$$

Since the above set is Borel, because the labelling $t(b)$ is Borel, it follows that the acceptance game is almsot a special case of a Muller-Borel game, with the only minor issue being that the arena is not strictly speaking tree shaped, since the play can stay in the same node for two consecutive positions. This minor issue will be resolved shortly.

Lemma 4.2. *A tree is accepted by a nondeterministic if and only if Eve has a winning strategy for the corresponding acceptance game.*

Proof. From an accepting run, we easily construct a winning strategy for Eve which follows this run, and vice versa. The only issue that requires some attention is that when we convert a winning strategy into a run, we need to ensure that the states in the branches are defined in a Borel way. Indeed, consider a winning strategy σ for Eve, and for a branch $b \in 2^\omega$ choose $q \in Q$ to be the least state (in some chosen linear order on the states) which witnesses that b belongs to the set from (1), in the unique play of the game that is consistent with σ and which traces branch b. The map $b \mapsto q$ defined this way is Borel, since it is defined in terms of a Muller condition on the states visited infinitely often in the branch. □

Thanks to the acceptance game, a tree is rejected if and only if player Adam has a winning strategy in the acceptance game. As we will show in the following lemma, this can be expressed by a nondeterministic automaton, provided that the winning strategy has finite memory, which will be ensured by Conjecture 1.2.

Lemma 4.3. *Assume Conjecture 1.2. Then languages recognised by nondeterministic Muller automata on Borel trees are closed under complementation, and the complement automaton can be effectively computed.*

Proof. Fix a nondeterministic Muller automaton on Borel trees $\mathcal{A}$. We want to compute an automaton, of the same kind, for its complement language. This automaton will guess a finite memory winning strategy for Adam in the corresponding acceptance game. Such a strategy will exist thanks to the conjecture, as shown in the following claim.

Claim 4.4. *Assume Conjecture 1.2. Based on the state space Q of $\mathcal{A}$ one can compute a memory bound m such that for every input tree, the corresponding acceptance game can be won using a strategy with at most m memory states.*

Proof. This is esentially the content of the conjecture, with the minor issue that the games from the conjecture require that one always moves to a child, while the acceptance game stays for two positions in the same node: after choosing a state, and then after choosing a transition. This can be easily resolved by considering a new game, in which game always moves to the left child after selecting a transition, and the Borel part is suitably updated to take into account these artificial moves (i.e. one ignores every second direction in the branch). □

Let $M = \{1, \ldots, m\}$ be the memory that is needed according to the above claim. By the claim and determinacy of Muller-Borel games, it follows that an input tree t is rejected by the automaton $\mathcal{A}$ if and only if there exists a winning strategy of player Adam in the acceptance game that uses memory M. Such a strategy can be viewed as a function which assigns two pieces of information for each node $v \in 2^*$ in the input tree, namely the functions

$$\underbrace{\delta_v : M \times (\Delta_* + Q) \to M}_{\text{memory update}} \quad \text{and} \quad \underbrace{\sigma_v : M \times \Delta_* \to \{0, 1\}}_{\text{Adam's choice}} .$$

A nondeterministic automaton can guess such a labelling, since such a labelling can be thought of as a tree over alphabet

$$\Gamma \quad \stackrel{\text{def}}{=} \quad (M \times (\Delta_* + Q) \to M) \times (M \times \Delta_* \to \{0, 1\}) \ + \ \{\omega\} ,$$

where the label ω is used for branches. We will show that once an input tree and a strategy for Adam are given in this representation, then a nondeterministic (in fact deterministic) automaton can check if the strategy is winning for Adam. In other words, the following language, which we call the *strategy language*, is recognised by a nondeterministic automaton (in fact, a deterministic one will be enough):

$$\left\{ t \in \mathsf{T}(\Sigma \times \Gamma) \mid \begin{array}{l} \text{the } \Gamma \text{ part represents a winning strategy for} \\ \text{Adam in the acceptance game for the } \Sigma \text{ part} \end{array} \right\}.$$

This will complete the complementation proof, since by projecting the strategy language to the Σ part, we will get the complement of the language accepted by $\mathcal{A}$. To define an automaton for the strategy language, we observe that it only checks a certain ω-regular all branches. This is because a tree belongs to the strategy language if and only if: (*) for every branch $b \in 2^\omega$, if we take any play of the acceptance condition that is consistent with Adam's strategy and which stays on the branch b, then this play is winning for player Adam. To determine whether a play is winning for Adam, we need access to the label of the branch b in the input tree, since this label is used in the winning conditon. Once we know this label, condition (*) becomes an ω-regular property of the branch, since it can be defined in MSO over ω-words. Therefore, to show that the strategy language is recognised by an automaton, we can use the following claim, where the alphabet Σ is the alphabet $\Sigma \times \Gamma$ of the strategy language.

Claim 4.5. *Let Σ be a finite alphabet, and suppose that for each letter $a \in \Sigma$ we have an associated ω-regular language L_a over the alphabet $\Sigma \times \{0, 1\}$. Then there is a nondeterministic (even deterministic) Muller automaton over Borel trees which accepts a tree $t \in \mathsf{T}\Sigma$ if and only if for every branch $b \in 2^\omega$ with label a, the language L_a contains the ω-word given by the sequence*

$$(\textit{label of } t \textit{ in } i\textit{-th node on } b, \textit{direction in } i\textit{-th step of } b)_{i=1}^{\omega}.$$

Proof. For each language L_a take deterministic Muller automata on ω-words that recognises, and run these automata on in parallel (using a product construction) along each branch. The acceptance condition of the resulting tree automaton checks if the label of the branch is consistent with the states of the automata that appear along this branch. □

□

Let us now put together all results proved above to show that Conjecture 1.2 imlies decidability of the Borel MSO theory of $2^{\leq\omega}$, as stated in Shelah's Conjecture A.

Theorem 4.6. *Assume Conjecture 1.2. Then nondeterministic Muller automata on Borel trees define the same languages as Borel MSO, and translations in both directions are effective. Furthermore, the Borel MSO theory of $2^{\leq\omega}$ is decidable.*

Proof. To go from an automaton to a formula of Borel MSO, one simply uses the logic to express the existence of an accepting run, in the usual way. To go in the opposite, one uses the closure properties that have been established above: intersections and unions (Lemma 3.4), projections (Lemma 3.5) and complementation (Lemma 4.3). The conjecture is used only for complementation. Finally, to decide the Borel MSO theory, we convert a formula into an equivalent automaton, and then use Lemma 3.3 to check if it is nonempty. □

4.3 Boolean combinations of F_σ sets

In Borel MSO, we have the same syntax as for MSO, except that the semantics are changed so that set quantification is restricted to Borel sets. One could go further in this direction and further restrict quantification to finite Boolean combinations of F_σ-sets. We recall that F_σ-sets are countable unions of open sets. The following theorem shows that, assuming our conjecture, the new logic is the same as Borel MSO, i.e. the same sentences are true (the two logics have the same syntax).

Theorem 4.7. *Assume Conjecture 1.2. Consider the variant of Borel MSO in which the same syntax is used, but the semantics are changed so that quantifiers range over Boolean combinations of F_σ sets. The new logic has the same true sentences as Borel MSO.*

Proof. Let us define $\mathsf{S}\Sigma \subseteq \mathsf{T}\Sigma$ to be the set of Borel trees where the labelling restricted to Boolean combinations of F_σ sets, i.e. for each label $a \in \Sigma$, its inverse image is a Boolean combination of F_σ sets. We will prove that for every formula MSO φ that describes a property of trees labelled by an alphabet Σ, we have

$$\underbrace{t \models_1 \varphi}_{\text{Borel semantics}} \quad \Leftrightarrow \quad \underbrace{t \models_2 \varphi}_{F_\sigma \text{ semantics}} \qquad \text{for every tree } t \in \mathsf{S}\Sigma. \tag{2}$$

The proof is a straightforward induction, and the only interesting case is that of a quantifier, say an existential one. To prove this step, we convert the quantified formula into an automaton using Theorem 4.6, and then we apply the following claim to show that the quantified set can always be chosen so that it is a Boolean combination of F_σ sets.

Claim 4.8. *Let $\mathcal{A}$ be a nondeterministic Muller automaton over Borel trees with input alphabet Γ, and let $f : \Gamma \to \Sigma$ be some projection function. For every tree $t \in \mathsf{T}\Sigma$, the following conditions are equivalent:*

(1) $\mathcal{A}$ accepts some $s \in \mathsf{S}\Gamma$ with $f(s) = t$;
(2) $\mathcal{A}$ accepts some $s \in \mathsf{T}\Gamma$ with $f(s) = t$;

Proof. The tree before the projection can be seen as an accepting run, and therefore in order to prove the claim it is enough to show that if an automaton $\mathcal{A}$ accepts a tree in $\mathsf{S}\Sigma$, then this can be done by an accepting run in $\mathsf{S}Q$. This follows from the acceptance game, since a run which arises from a finite memory strategy in the acceptance game will be necessarily be in $\mathsf{S}Q$. □

□

We can recast the above theorem, or more precisely the slightly stronger statement in (2), as follows. We can view Borel MSO as the first-order theory of the structure where the universe consists of Borel subsets of $2^{\leq\omega}$. Then (2) tells us the if we further restrict the universe to Boolean combinations of F_σ sets, then we get an elementary substructure. As we have remarked in the introduction, this type of situation does arise in the weaker setting of Borel MSO on 2^ω, and therefore it is plausible that it could also occur for $2^{\leq\omega}$.

5 Prioritized Borel games

As we have explained in the previous section, the crucial point of our approach is finite memory determinacy for acceptance games that arise from alternating automata. The rest of this paper is devoted to a study of these games, and evidence for the conjecture that they are determined with finite memory.

5.1 Definition and finite memory conjecture

Before continuing, we clean up the definition of the games, by removing certain artefacts that arise from automata and which are not important to understand the games. The cleaned up version is presented below; one of its advantages is that it uses priorities (similar to the parity condition) instead of a Muller condition.

Definition 5.1 (Prioritized Borel game)**.** *A* prioritized Borel game *is defined in the same way as a Muller-Borel game, except that the winning condition in item 2 of Definition 1.1 is replaced by:*

(2) ***Prioritized Borel condition.*** *The winning condition is described using:*
   (a) a priority function, which assigns a priority *from* $\{1, ..., k\}$ *to every edge in the game, for some* $k \in \{1, 2, \ldots\}$*; and*
   (b) a sequence of Borel sets

$$W_1, \ldots, W_k \subseteq 2^\omega.$$

The winner of an infinite play is obtained as follows: take i to be the maximal priority that occurs infinitely often on the edges of the play. Then, Eve wins if the branch part of the play belongs to W_i*, and Adam wins otherwise.*

Note that the tree-like structure influences the information that is explicitly available for the players during a play. Namely, the sequence of produced nodes of the tree is encoded in the arena, while the sequence of states or priorities is not; players need memory to keep track of this latter information.

In the remaining of this paper we discuss the following conjecture.

Conjecture 5.2. *For every prioritized Borel game, one of the players has a winning strategy that uses finite memory. Furthermore, an upper bound on the memory size can be computed based on the number of states and number of priorities.*

We note that prioritized Borel games are determined, that is, there is a "*winner*", as they are a special case of games with Borel winning conditions. The interesting part of the conjecture is therefore the one concerning memory.

Remark 5.3. The above conjecture implies that, for fixed number of states and priorities, there is a uniform bound on the memory needed. That is, for all integers n and k, there is a constant $c_{n,k}$ such that in any prioritized Borel game with $|Q| \leq n$ and using k priorities, one of the players has a winning strategy that uses at most $c_{n,k}$ memory states.

Indeed, if this was not the case, there would be a sequence of games where the the winner (that we can assume to be Eve by symmetry) requires unbounded memory. We could combine these games in a single one: Adam can stay in the leftmost branch and move into the ith game at the ith node. If he stays in the leftmost branch forever, he loses.

Example 5.4 (One state and priority). The get a feeling for the conjecture, let us discuss the special case where there is one state and one priority. In this case, the finite memory assumption trivialises, so the above conjecture is just a restatement of Borel determinacy. Indeed, in this case there is no difference between general strategies and finite memory strategies (even memoryless strategies), since the current position stores the entire history of the play.

Next, we show an example of a prioritized Borel game where the winner requires memory.

Example 5.5 (Copying game). In Figure 1 we show a prioritized Borel game with only one state per tree node, and using two priorities. In this game, Adam chooses to produce a priority 1 or 2, and then Eve chooses to go left or right in the tree. We let the Eve-winning Borel subsets be W_1 = branches that eventually only take the left child, and W_2 = branches that contain a right child infinitely often. That is, Eve wins if she goes right infinitely often if and only if Adam produces 2 infinitely often.

Eve can win this game: each time that Adam takes a 2-edge, she can go to the right child in the tree. However, she needs memory to implement this strategy. In fact, she cannot win with a memoryless strategy, as such a strategy determines one branch, and Adam can produce only priorities 1 or 2 depending on whether right children appear infinitely often in the branch.

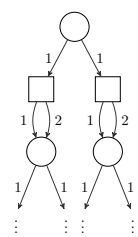

Figure 1: The prioritized Borel game from Example 5.5. Eve nodes are represented as circles, and Adam nodes as squares.

5.2 Upper bounds: Finite-memory results

The rest of this paper is dedicated to a discussion of Conjecture 5.2. In this section, we give some positive evidence, by proving some special cases of the conjecture. In the next section, we exhibit some challenges in proving the conjecture, in particular we will show that unbounded memory is necessary.

5.2.1 One priority. We start discussing the case of prioritized Borel games with just one priority. Although this case is relatively simple, it illustrates well our general strategy for tackling Conjecture 5.2. In this case, there is only one Borel winning set W_1 and the only condition for Eve to win is that the final branch lies on this set.

Proposition 5.6. *In every prioritized Borel game with one priority, the winner has a memoryless strategy.*

Proof. Consider a game with one priority, and let W_1 be Eve's winnning set of branches. Assume that she has a winning strategy σ, and let $W_\sigma \subseteq 2^\omega$ be the subset of branches that can be produced when Eve follows the strategy σ.

We claim that W_σ is a topologically closed set. Indeed, W_σ contains the branches avoiding the subtrees under nodes in

$$N = \{w \in 2^* \mid w \text{ is unnatainable using } \sigma\},$$

that is, it is the complement of a union of cylinders. Also, $W_\sigma \subseteq W_1$, as the strategy is winning.

Therefore, in the game obtained by replacing the winning set W_1 by W_σ, Eve also has a winning strategy (namely, σ). In this game, the winning condition is closed, and therefore Eve has a memoryless winning strategy: she just avoids edges entering N. □

In the proof above, we achieved our goal by replacing the set W_1, of arbitrary complexity in the Borel hierarchy, by a topologically simple one: a closed set, for which memoryless strategies suffice.

More generally, Conjecture 5.2 holds for Borel subsets of "low" complexity. If all winning sets $W_1, \dots, W_k$ are Boolean combination of at most m F_σ-sets, then the winner has a strategy that uses memory bounded by a constant that only depends on k and m (this follows from the classical finite-determinacy result of Büchi-Landweber [1] generalized to infinite arenas [**?**]).

We conjecture that in every prioritized Borel game, we can replace the winning sets $W_1, \dots, W_k$ by smaller subsets that are Boolean combination of F_σ-sets of bounded complexity (depending on the number of states and priorities) while preserving the winner. This implies Conjecture 5.2. In fact, both conjectures are equivalent: given a finite-memory strategy, we can replace the winning sets by the sets that are consistent with the strategy (just as in the proof of Proposition 5.6 above); these sets are of bounded topological complexity.

5.2.2 Two priorities. We prove Conjecture 5.2 in the case of two priorities. Example 5.5 uses two priorities, so we know that memoryless strategies do not suffice in this case.

Proposition 5.7. *In every prioritized Borel game with two priorities, the winner has a strategy using two states of memory.*

Proof. Assume that Eve wins a given prioritized Borel game with two priorities using a strategy σ. Intuitively, we will identify some tree nodes, that we call *witnesses*, with the property that the strategy σ produces priority 2 between two of them. This information is then encoded explicitly in the arena, and a finite-memory strategy can combine choices of σ with this extra information.

Witness nodes are formally defined recursively. The root is a witness. For a node v, let w be the last ancestor that is a witness. Then v is a witness if in any play compatible with σ, there is an occurrence of priority 2 between v and w. (Note that witnesses are just nodes in the tree-part of the arena 2^*, not positions).

We define a new strategy using two states of memory. The memory will recall whether priority 2 has been produced since the last witness. We associate to each position (w, q) two histories. The new strategy will mimic the choice of σ after one of these histories, according to the information stored in the memory state. In particular, all attainable positions following this strategy are also attainable following σ.

Let $h_{\mathsf{maxlex}}(w, q)$ be a σ-consistent history leading to (w, q) maximizing the occurrences of priority 2 according to the lexicographic order. That is, it contains the history where priorities 2 occur as soon as possible, and as many as possible. Let $h_{\mathsf{no2}}(w, q)$ be any σ-consistent history leading to (w, q) that does not contain any occurrence of 2 since the last witness. (If w is a witness, this is just any σ-consistent history). Now, assume that in a play we are in a position (w, q). If priority 2 has been produced since the last witness, then Eve follows the σ-choice after $h_{\mathsf{maxlex}}(w, q)$. On the contrary, she follows the σ-choice after $h_{\mathsf{no2}}(w, q)$. We show that this strategy is winning.

Let π be an infinite play consistent with the new strategy, laying in a branch $b \in 2^\omega$. Let π_n be its prefix of length n, and h_n be the history considered at that point in the definition of the strategy (that is, h_n equals $h_{\mathsf{maxlex}}(w, q)$ or $h_{\mathsf{no2}}(w, q)$ for the node (w, q) reached by π_n). By König's lemma [**?**], there is an infinite play h_∞ consistent with σ and such that any prefix of it is a prefix of some h_n. In particular, h_∞ lies on the branch b. We distinguish three cases.

First, assume that b contains infinitely many witnesses. In this case, we claim that π contains 2 infinitely often and $b \in W_2$. Indeed, between any two witnesses 2 appears at least once in π, as otherwise Eve would extend a σ-consistent history avoiding 2 in this interval, contradicting the definition of witness. By definition of witness, h_∞ also contains 2 infinitely often. Thus, $b \in W_2$ and Eve wins.

Assume now that b contains finitely many witnesses and priority 2 appears infinitely often in π. In particular, 2 appears after the last witness. After such an occurrence of 2, all considered histories are of the form $h_n = h_{\mathsf{maxlex}}(w, q)$. Thus, by definition of h_{maxlex}, the prefixes of the sequences of priorities in h_n converge to a profile in $\{1, 2\}^*$. That is, the sequence of priorities in h_∞ is the well-defined limit of the profiles of h_n. As π contains 2 infinitely often, every history h_n can be extended with a σ-consistent run having a further occurrence of 2. Since h_n maximizes the number of occurrences of 2 (lexicographically) and the prefixes stabilise, these prefixes will contain an unbounded number of 2s. Therefore, h_∞ contains 2 infinitely often, so $b \in W_2$ and Eve wins.

Finally, assume that π contains 2 finitely often. We show that there is a σ-consistent run laying in b with finitely many 2, so $b \in W_1$. Let n_0 be the position of the last witness in b. By definition of witness, for every $n > n_0$ there is a σ-consistent history h'_n extending h_{n_0} with no occurrence of 2 after n_0. König's lemma yields an infinite σ-consistent play in b avoiding 2 after n_0, as desired. □

Remark 5.8. We discuss how the above proof relates to the topological simplication approach described above.

The definition of witness yields a coloring of the tree with two colors ("witness" or "not a witness"). We can replace W_1 by the set of branches that are compatible with σ and contain finitely many witnesses, which is a F_σ-set contained in W_1.

The corresponding subset of W_2 is slighlty more complex to define; it can be described as the intersection of σ-compatible branches with the union of two sets, corresponding to the first two cases examined in the above proof. The first one is the set of branches that contain infinitely many witnesses (a G_δ set, i.e., the complement of a F_σ). The second is the set of branches in which the limit of the h_{maxlex} σ-histories contains infinitely many 2s. This is also a G_δ.

This provides two topologically simple subsets, and the proof above describes a winning strategy for Eve in the obtained game.

In the next section, we will see that in games with 3 priorities there is not a constant bound in the memory required, independent of the number of states.

5.2.3 One state. We present a result that is orthogonal to Proposition 5.7: a proof of Conjecture 5.2 for the case of games with a single state. Note that, as priorities in prioritized Borel games appear on transitions, despite the single-state restriction, a player my have up to $2k$ options in each move.

Proposition 5.9. *In every prioritized Borel game with a single state and k priorities, the winner has a strategy using at most 2^{k-1} states of memory.*

Proof. □

5.2.4 Alternatingly nested sets. We say that a sequence of subsets $W_1, \dots, W_k \subseteq 2^\omega$ is *alternatingly nested* if for every $1 \leq i \leq k$,

$$\text{either} \quad W_i \subseteq \bigcap_{j<i} W_j, \quad \text{or} \quad W_i \supseteq \bigcup_{j<i} W_j.$$

Proposition 5.10. *If the Borel sets of a prioritized Borel game are alternatingly nested, then one of the players can win using a memoryless strategy.*

5.3 Lower bounds

In this section we prove some lower bounds on the memory required in prioritized Borel games. Namely, we see that there are no uniform bounds on neither the number of priorities nor the number of states of the games. We hope that this will help on guiding future approaches to solving Conjecture 5.2.

In this section, we will use prioritized Borel games where the branching of the tree may be greater than 2, but still finite and bounded. Formally, we let C be a finite set of children, and the set of positions is a subset of $C^* \times Q$. These games can be simulated by a binary branching prioritized Borel game: it suffices to introduce $\log(\lceil |C| \rceil)$ intermediate levels between each two nodes of the tree of a given C-branching game. This operation does not add priorities, nor states.

We recall that we use circles to represent vertices owned by Eve, and squares to represent those owned by Adam.

5.3.1 Many priorities, one state. We generalize Example 5.5 to games with k priorities, for any k, and show that in these games Eve requires k states of memory to win.

Proposition 5.11. *For every k, there is a prioritized Borel games with a single state and k priorities where Eve requires k states of memory to win.*

Proof. Consider the (k-branching) prioritized Borel game using k priorities represented in Figure 2. In this game, Adam picks a priority in $\{1, \dots, k\}$, then, Eve chooses to go to one of k possible children (that we consider ordered following the numbering). Eve wins if the highest priority that appears infinitely often coincides with the highest children that is taken infinitely often. Formally, W_i is the set of branches where the ith children is taken infinitely often, but for all $j > i$, the jth children is taken only finitely often.

Eve can win this game: she just needs to copy Adam's move (go to the ith children if Adam picks priority i). This strategy can be implemented with k memory states, using the states to record the priority played by Adam.

We claim that this is optimal. Assume by contradiction that Eve has a winning strategy using $k-1$ memory states. Intuitively, in every step she confounds two priorities; we will use this fact two build two plays on the same branch that produce a different maximal priority infinitely often. For each level n of the tree, we define by recursion:

- a position of the game (controlled by Adam when n is even, and by Eve when n is odd),
- a memory state m_n, and
- two priorities, i_n, i'_n (only for odd n).

We let v_0 be the root of the tree[1] and m_0 the initial memory state. Assume that these elements have been defined up to level n, for n even (v_n controlled by Adam). As there are only $k-1$ memory states, there are two priorities, i_{n+1} and i'_{n+1}, such that the memory update from state m_n when reading these two priorities goes to the same memory state: we let m_{n+1} be this state. We let v_{n+1} be the only successor of v_n and v_{n+2} be the child of v_{n+1} picked by the strategy in the memory state m_{n+1}. Finally, let m_{n+2} be the memory state obtained by updating m_{n+1} over this last transition.

This shows that, when Eve plays according to this strategy, Adam can decide to produce a play in the branch $v_0 v_1 v_2 \dots$, while producing any sequence of priorities in $(i_0 + i'_0)(i_1 + i'_1) \dots$. Taking the sequences of the maximal and minimal elemets of $\{i_n, i'_n\}$, respectively, we obtain two sequences with different largest priority, concluding the proof. □

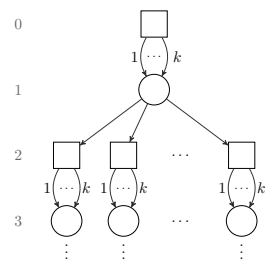

Figure 2: The prioritized Borel game from Proposition 5.11.

5.3.2 Three priorities, many states. We have seen that for games using two priorities we can obtain a memory bound independent of the number of states of the game (Proposition 5.7). We show that this is no longer the case for games using three priorities.

Proposition 5.12. *For every s, there is a prioritized Borel game using 3 priorities where Eve requires s memory states to win.*

We describe a prioritized Borel game G_s with $2s$ states, 3 priorities and tree-branching $2s$ (see also Figure **??**). That is, the set of children is $C = \{c_1, \dots, c_{2s}\}$, that we order following the numbering. In each tree node there are s states controlled by Adam: $q_1, \dots, q_s$; and s states controlled by Eve: $p_1, \dots, p_s$. When in a state q_i, Adam can choose between:

- go to any child between c_1 and c_{2i}, to the state q_i of that node, and producing priority 1, or
- move to state p_i in child c_{2i-1}, producing priority 1.

When in a state p_i, Eve can choose to either produce priority 2 and go to the child c_{2i}, or to produce priority 3 and go to the first child c_1. In both cases, we go to the state q_1

[1] As this is a one-state game, we confound positions of the game and nodes of the tree.

The winning condition is as follows. The set W_1 is full (if eventually only priority 1 is produced, Eve wins). The set W_2 is empty (Adam always wins if 2 is produced infinitely often an 3 finitely often). If 3 is produced infinitely often, Eve wins if and only if the highest children that appears infinitely often is even.

We give an equivalent description as a game alternating turns: First, Adam picks a number $i \leq s$ and produces a sequence in $\{c_1, \ldots, c_{2i}\}^* c_{2i-1}$. Then, he hands over to Eve, who can choose to append c_{2i} to the sequence, or to "emit a signal" (this corresponds to produce priority 3 above). Then, we start all over again. In order to win, Eve needs to both emit a signal infinitely often, and ensure that the largest element in the sequence produced infinitely often is even.

Eve can win the game by alternating her choices each time we enter the state p_i. That is, if we enter a state p_i for the nth time, she produces priority 2 and goes to c_{2i} if n is even, and produces priority 3 if n is odd.

Lemma 5.13. *Eve cannot win the game G_s using less than s memory states*

Proof. We prove the lower by induction on s. We will use a slightly stronger induction hypothesis: Eve cannot win G_s with a strategy that uses $s-1$ reachable memory states in positions owned by her (that is, positions of the form (w, p_i)).

For the base case, $s = 1$, a strategy using 1 memory state in Eve positions is just a memoryless strategy; we show that they do not suffice to win G_1. Fix a memoryless strategy, that is, in each node $w \in C^*$ fix Eve's choice of producing priority 3 or going to children c_2 from p_1. For every node $v \in C^*$, there is some descendent w in the leftmost branch below v such that Eve goes to c_2 from (w, p_1), as otherwise Adam can produce an infinite play in this branch. But then, Adam can let Eve play only in these nodes, so priority 3 will never be produced. Therefore, the strategy is not winning.

Now, assume by contradiction that Eve has a winning strategy σ in G_s using $s-1$ memory states in her positions. We can rename these memory states so that they are in a same set M of size $s-1$. We say that Eve *plays c_{2i} unconditionally* from a position (w, p_i), if from every memory state the strategy σ choses to play c_{2i} and priority 2 from the position (w, p_i). We let $C_{s-1} = \{c_1, \ldots, c_{2s-1}\}$. We say that game position (w, p_i) is a *C_{s-1}-descendant* of a node w_0 if $w \in w_0 C_{s-1}^*$, that is, w is below w_0, and the child c_{2s} does not appear after w_0.

Claim 5.14. *There exists a node such that for any of its C_{s-1} descendants (w, p_i), Eve does not play c_{2i} unconditionally from (w, p_i).*

Proof. Assume that this is not the case. Then, we can build by induction a play consistent with σ in which priority 3 is never produced: In each step, Adam choses a descendant of the current node where Eve plays c_{2i} unconditionally. He can go to that position, and let her play, producing c_{2i} and priority 2. Repeating this strategy, Adam produces an infinite path where priority 3 is never visited, so it is losing for Eve, a contradiction. □

Since at the start of the play Adam may go to any node of the tree, we may assume that the initial node given in the previous claim is the root of the tree. We fix a function

$$f \colon C_{s-1}^* \times \{p_1, \ldots, p_s\} \to M$$

that associates to each position (w, p_i) a memory state where the strategy σ choses to produce priority 3, instead of going to c_{2i}. (This function is well-defined by the previous claim.)

The induction hypothesis is used to obtain the next claim.

Claim 5.15. *Let $w_0 \in C_{s-1}^*$. There is a C_{s-1}-descendant (w, p_i) of w_0 and a σ-play going from (w_0, q_1) to (w, p_i) such that the memory state when in (w, p_i) is $f(w, p_i)$.*

Proof. Assume that this is not the case, and consider the restriction to the game G_{s-1} starting at the node w_0. In each position (w, p_i) controlled by Eve, there is a memory state that is not reachable. Therefore, Eve wins G_{s-1} using only $s-2$ memory states in her positions, contradicting the induction hypothesis. □

To conclude, we build recursively an infinite play where Eve never choses to pick c_{2i}; such a play is losing for her. At every step, it suffices that Adam plays accordingly to the play given by the previous claim. □

We have not succeed on proving Conjecture 5.2 for games with 3 priorities.

6 Conclusions

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A Appendix A: